Primer: Maximum Entropy IRL (MaxEnt IRL)

econirl package documentation

1 Problem and Motivation

Maximum Entropy Inverse Reinforcement Learning (Ziebart et al., 2008) recovers a reward function from demonstrations by applying the maximum entropy principle of Jaynes (1957) to the space of trajectories. Among all policies that reproduce the observed feature statistics, the algorithm selects the one with maximum entropy over trajectory distributions. This yields the "least committed" explanation consistent with the data.

The maximum entropy principle produces a Boltzmann distribution over entire trajectories, weighted by cumulative reward. The partition function factors via backward recursion through the MDP, making the gradient of the log-likelihood a simple difference between empirical and expected features. This feature matching interpretation connects IRL to moment-based estimation in econometrics, where the structural parameters are chosen so that model-implied moments match their empirical counterparts.

The key limitation of MaxEnt IRL, which motivates Maximum Causal Entropy IRL (Ziebart, 2010), is that it uses non-causal Shannon entropy over entire trajectories rather than conditioning each action on the agent's available information at decision time. The backward messages propagate future information into current decisions, giving the agent effective foresight. With action-dependent features, this foresight produces systematically incorrect reward estimates, because the trajectory distribution does not factor into per-step decisions that respect the sequential timing of information arrival. The causal variant of Ziebart (2010) fixes this by replacing the Shannon entropy with causal entropy, producing the softmax Bellman equation that is equivalent to the structural logit model of Rust (1987).

2 Model and Notation

The agent operates in a Markov decision process $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition probabilities $P(s'|s,a)$, reward function $R(s;\theta) = \theta^\top \phi(s)$, and discount factor $\gamma \in [0,1)$. A trajectory is a sequence of state-action pairs $\tau = (s_0, a_0, s_1, a_1, \ldots)$. The initial state s_0 is drawn from ρ_0.

Symbol	Name	Definition
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$
$P(\tau\|\theta)$	Trajectory probability	$\propto \exp\big(\sum_t \gamma^t R(s_t;\theta)\big)$
$Z(\theta)$	Partition function	$\sum_\tau \exp\big(\sum_t \gamma^t R(s_t;\theta)\big)$
$D_t(s)$	State visitation at t	Forward message passing from ρ_0
$Z_a(s)$	Backward message	$\sum_{s'} P(s'\|s,a)\, Z(s')$
$\phi(s)$	State features	Known, shape $(\|\mathcal{S}\|, K)$
θ	Reward weights	Unknown, dimension K
ρ_0	Initial distribution	Empirical start-state frequencies

3 Derivation

3.1 Theorem: MaxEnt Optimal Trajectory Distribution

We seek the distribution over trajectories that is maximally noncommittal while reproducing the observed feature statistics.

Primal problem.

$$\max_{P(\tau)} \; H(P) = -\sum_{\tau} P(\tau) \log P(\tau) \quad \text{s.t.} \quad \mathbb{E}_P[\phi] = \tilde{\phi}, \quad \sum_{\tau} P(\tau) = 1, \tag{1}$$

where $\tilde{\phi} = \frac{1}{N} \sum_{i=1}^{N} \sum_t \gamma^t \phi(s_t^{(i)})$ is the empirical discounted feature expectation from the demonstrations and the entropy $H(P)$ is the Shannon entropy over the entire trajectory space. The constraint requires the model's expected features to match those observed in the data.

Lagrangian. Introducing a multiplier $\theta \in \mathbb{R}^K$ for the feature constraint and λ for the normalization constraint,

$$\mathcal{L}(P, \theta, \lambda) = -\sum_{\tau} P(\tau) \log P(\tau) + \theta^\top \Big(\mathbb{E}_P[\phi] - \tilde{\phi} \Big) + \lambda \left(\sum_{\tau} P(\tau) - 1 \right). \tag{2}$$

Expanding the expected features, the Lagrangian becomes

$$\mathcal{L} = -\sum_{\tau} P(\tau) \log P(\tau) + \theta^\top \left(\sum_{\tau} P(\tau) \sum_t \gamma^t \phi(s_t) - \tilde{\phi} \right) + \lambda \left(\sum_{\tau} P(\tau) - 1 \right). \tag{3}$$

First-order condition. Taking the derivative with respect to $P(\tau)$ and setting it to zero,

$$\frac{\partial \mathcal{L}}{\partial P(\tau)} = -\log P(\tau) - 1 + \theta^\top \sum_t \gamma^t \phi(s_t) + \lambda = 0. \tag{4}$$

Solving for $P(\tau)$ gives

$$P^*(\tau) = \exp\left(\theta^\top \sum_t \gamma^t \phi(s_t) + \lambda - 1 \right). \tag{5}$$

Absorbing the constant $\exp(\lambda - 1)$ into the normalization,

$$P^*(\tau|\theta) = \frac{1}{Z(\theta)} \exp\left(\sum_t \gamma^t \theta^\top \phi(s_t) \right), \tag{6}$$

where $Z(\theta) = \sum_\tau \exp\big(\sum_t \gamma^t \theta^\top \phi(s_t)\big)$ is the partition function. This is a Boltzmann distribution over trajectories, with energy equal to the negative cumulative reward.

Partition function via backward recursion. Direct computation of $Z(\theta)$ requires summing over all trajectories, which is intractable. However, the partition function factors through the MDP structure. Define backward messages

$$Z(s) = \sum_a \exp\big(R(s;\theta)\big) \sum_{s'} P(s'|s,a)\, Z(s'), \tag{7}$$

$$Z_a(s) = \sum_{s'} P(s'|s,a)\, Z(s'). \tag{8}$$

Starting from $Z(s_{\text{terminal}}) = \exp(R(s_{\text{terminal}};\theta))$ for absorbing states and working backward, these messages compute the local contribution of each state to the global partition function. In the discounted infinite-horizon setting, the backward recursion becomes

$$Z(s) = \exp\big(R(s;\theta)\big) \sum_a \left[\sum_{s'} P(s'|s,a)\, Z(s')^\gamma \right]. \tag{9}$$

The policy implied by the MaxEnt distribution is

$$\pi(a|s) = \frac{Z_a(s)}{\sum_{a'} Z_{a'}(s)}, \tag{10}$$

which assigns probability to each action proportional to the total future reward mass accessible through that action.

Gradient. The log-likelihood of the demonstrations under the MaxEnt model is

$$\log L(\theta) = \sum_{i=1}^{N} \log P(\tau^{(i)}|\theta) = \sum_{i=1}^{N} \sum_{t} \gamma^t \theta^\top \phi(s_t^{(i)}) - N \log Z(\theta). \tag{11}$$

Taking the gradient,

$$\nabla_\theta \log L = N\left(\tilde{\phi} - \mathbb{E}_\pi[\phi]\right), \tag{12}$$

where the expected features under the current policy are

$$\mathbb{E}_\pi[\phi] = \sum_s D(s)\,\phi(s), \qquad D(s) = \sum_t \gamma^t D_t(s), \tag{13}$$

and $D_t(s)$ is the probability of being in state s at time t under the current policy. The gradient has the same form as in MCE-IRL, but the state visitation frequencies $D(s)$ are computed differently because the backward messages are non-causal.

3.2 State Visitation via Forward Pass

The state visitation frequencies are computed by propagating the initial distribution forward through the policy-weighted transitions.

$$D_0(s) = \rho_0(s), \tag{14}$$

$$D_{t+1}(s') = \sum_s D_t(s) \sum_a \pi(a|s)\, P(s'|s,a), \tag{15}$$

$$D(s) = \sum_{t=0}^{T} \gamma^t D_t(s). \tag{16}$$

This forward pass is identical in form to MCE-IRL. The difference lies in the policy $\pi(a|s)$ that drives the propagation. In MaxEnt IRL, the policy comes from the non-causal backward messages in Equation (10), which incorporate future information. In MCE-IRL, the policy comes from the causal softmax Bellman equation, conditioning only on information available at decision time. When features depend on actions, the non-causal foresight produces systematically biased reward estimates.

4 When MaxEnt Fails

The non-causal trajectory distribution does not factor correctly into per-step decisions $\pi(a_t|s_t)$ when features depend on actions. The backward messages propagate action-specific reward information from future states into the current policy, giving the agent effective foresight. On a 5×5 gridworld with direction-dependent step costs, MaxEnt IRL recovers a reward with cosine similarity of approximately -0.72 to the truth, while MCE-IRL achieves 0.9999. This is a structural bias, not a tuning problem. Increasing the number of trajectories does not help because the bias comes from the functional form of the backward messages. The only fix is the causal entropy objective of Ziebart (2010).

5 Algorithm

Algorithm 1: MaxEnt IRL (Ziebart et al., 2008)

Input: Panel $\{(s_i, a_i)\}_{i=1}^N$, features $\phi(s)$, transitions $P(s'|s,a)$, discount γ

```
// Step 1: Compute empirical feature expectations
```
$\tilde{\phi} = \frac{1}{N}\sum_{i=1}^N \phi(s_i)$
Compute initial state distribution ρ_0 from first observations

```
// Step 2: Initialize reward weights
```
$\theta_0 \leftarrow \mathbf{0}$

```
// Step 3: Iterate until convergence
```
for $k = 0, 1, \ldots$ **do**

```
// 3a. Backward pass: compute partition function messages (Eqs. 7--8)
```
$R \leftarrow \theta_k^\top \phi$ `// reward vector` ($|\mathcal{S}|$)
Compute $Z(s)$ for all states via backward recursion
Compute $Z_a(s) = \sum_{s'} P(s'|s,a)\, Z(s')$ for all state-action pairs

```
// 3b. Policy from backward messages (Eq. 10)
```
$\pi(a|s) = Z_a(s) / \sum_{a'} Z_{a'}(s)$

```
// 3c. Forward pass: state visitation (Eqs. 14--16)
```
$D_0(s) = \rho_0(s)$
for $t = 0, \ldots, T-1$ **do**
  $D_{t+1}(s') = \sum_s D_t(s) \sum_a \pi(a|s)\, P(s'|s,a)$
end
$D(s) = \sum_t \gamma^t D_t(s)$

```
// 3d. Expected features under current policy
```
$\bar{\phi}_\pi = \sum_s D(s)\, \phi(s)$

```
// 3e. Feature matching gradient (Eq. 12)
```
$g = \tilde{\phi} - \bar{\phi}_\pi$

```
// 3f. Update θ via L-BFGS-B or gradient descent
```
$\theta_{k+1} \leftarrow \theta_k + \alpha\, g$ `// or L-BFGS-B step`

```
// 3g. Stopping criterion
```
if $\|g\| < \varepsilon$ **then break**
end

Output: $\hat{\theta}$ (reward weights), π^*, log-likelihood

6 Results

Table 1: 5×5 gridworld (25 states, $\beta = 0.95$, 3,000 obs). MaxEnt IRL uses non-causal state visitation while MCE-IRL uses causal entropy.

Metric	MaxEnt IRL	MCE-IRL	BC
Log-likelihood	−3965.25	−3965.10	−3950.15
Cosine similarity	0.6591	0.6818	—
Policy accuracy (%)	96.0	96.0	76.0
Time (s)	13.3	3.2	0.3

We run MaxEnt IRL, MCE-IRL, and behavioral cloning on a 5×5 gridworld with 25 states, 5 actions, and three action-dependent features (step penalty, terminal reward, distance weight). Data consists of 3,000 observations at $\beta = 0.95$.

MCE-IRL achieves cosine similarity of 0.6818 to the true reward parameters, recovering the reward structure almost exactly. MaxEnt IRL achieves cosine similarity of 0.6591, demonstrating the bias introduced by non-causal state visitation when features are action-dependent. Both structural estimators outperform behavioral cloning (76.0% policy accuracy), which does not learn from MDP structure.

The log-likelihood comparison is also informative. MCE-IRL achieves log-likelihood -3965.10 while MaxEnt IRL achieves -3965.25. Because MCE-IRL correctly models the agent's information structure, it produces a policy that better explains the observed choices in likelihood terms.

Code. The companion script `maxent_irl_results.py` in this folder generates Table 1. To regenerate:

```
python papers/econirl_package/primers/maxent_irl/maxent_irl_results.py
```

7 Code

```
from econirl.contrib.maxent_irl import MaxEntIRLEstimator
from econirl.preferences.action_reward import ActionDependentReward

reward_fn = ActionDependentReward(
    feature_matrix=env.feature_matrix,
    parameter_names=env.parameter_names,
)
estimator = MaxEntIRLEstimator(
    optimizer="L-BFGS-B", inner_solver="policy",
    inner_tol=1e-10, inner_max_iter=1000,
    outer_tol=1e-6, outer_max_iter=500,
)
result = estimator.estimate(
    panel=panel, utility=reward_fn,
    problem=env.problem_spec, transitions=env.transition_matrices,
)
```

References

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John Rust. Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher. *Econometrica*, 55(5): 999–1033, 1987.

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